Real Python What's New in Python 3.15

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Upgrading

- Python 3.15.0 arrives on October 1, 2026
- The new syntax is additive, but a few old APIs are gone
- Play with it today, let servers wait for 3.15.1
- Most compiled packages need new wheels first

Run 3.15 Without Changing Your Python

```
$ uv run --python 3.15 python
```

Confirm the Version You Run

```
>>> import sys
>>> sys.version_info[:2]
(3, 15)
```

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`python 3.15` · `uv` · `install python`

Lazy Imports

- `lazy` is a soft keyword, so it's not reserved
- The module loads on first use, not on import
- Module scope only: no functions, no `try`
- Import errors surface at first use, not at import
- Great for CLIs with slow, rarely used imports

Defer an Import Until First Use

```
lazy import json
lazy from pathlib import Path

print("Starting up...")  # Nothing loaded
data = json.loads('{"k": 1}')  # Loads now
```

Make Eligible Imports Lazy

```
$ python -X lazy_imports=all app.py
```

Opt In Without the Keyword

```
__lazy_modules__ = ["json", "pathlib"]

import json  # Lazy
import os    # Still eager
```

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`lazy imports` · `imports` · `startup time`

Unpacking in Comprehensions

- Use `*` and `**` directly in a comprehension
- Replaces nested loops and `chain.from_iterable()`

Flatten With a Star

```
rows = [[1, 2], [3, 4]]
[*row for row in rows]  # [1, 2, 3, 4]
```

Merge Dictionaries and Sets

```
dicts = [{"a": 1}, {"b": 2}]
{**d for d in dicts}  # {'a': 1, 'b': 2}

{*s for s in [{1, 2}, {3}]}  # {1, 2, 3}
```

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`comprehensions` · `unpacking` · `iterables`

Immutable Dictionaries

- `frozendict` is a new built-in type
- Hashable when its values are, so it can be a key
- It freezes the mapping, not the objects inside
- It's not a `dict` subclass, but it compares equal

Create a Read-Only Mapping

```
config = frozendict(host="localhost")
config["host"]        # 'localhost'
config["host"] = "x"  # TypeError
```

Use a Mapping as a Key

```
cache = {frozendict(a=1): "hit"}
cache[frozendict(a=1)]  # 'hit'
```

Learn More on realpython.com/search:

`frozendict` · `dictionaries` · `immutability`

Sentinel Values

- `sentinel` is a new built-in type
- Use it when `None` is a valid argument
- Each call returns a new object, so create it once
- Compare sentinels with `is`, never with `==`

Tell a Missing Value From None

```
MISSING = sentinel("MISSING")

def greet(name=MISSING):
    if name is MISSING:
        name = "world"
    return f"Hello, {name}!"
```

See Why You Create It Once

```
>>> MISSING
MISSING
>>> MISSING is sentinel("MISSING")
False
```

Learn More on realpython.com/search:

`sentinel` · `default arguments` · `none`

Profiling

- `profiling` collects the profilers in one namespace
- Tachyon samples stacks instead of tracing calls
- Low enough overhead for production processes
- `cProfile` lives on as `profiling.tracing`

Profile a Script

```
$ python -m profiling.sampling run app.py
```

Attach to a Running Process

```
$ python -m profiling.sampling attach 5171
```

Watch the Stacks Live

```
$ python -m profiling.sampling run \
      --live app.py
```

Learn More on realpython.com/search:

`profiling` · `performance` · `optimize code`

Smarter Error Messages

- Hints now cover method names from other languages
- Suggestions reach one level into an object

Get the Python Name for a Method

```
[1, 2, 3].push(4)
# AttributeError: 'list' object has no
# attribute 'push'. Did you mean
# '.append'?
```

See Through a Wrapper Object

```
container.area
# AttributeError: 'Container' object has
# no attribute 'area'. Did you mean
# '.inner.area' instead of '.area'?
```

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`error messages` · `attributeerror` · `debugging`

UTF-8 by Default

- Text I/O is UTF-8 whatever the locale says
- Still pass `encoding=` if the code runs on 3.14
- `encoding="locale"` opts back in per call
- `PYTHONUTF8=0` restores the old behavior

Open a File Without Guessing

```
with open("data.txt") as file:
    text = file.read()
```

Check the Default

```
>>> open("data.txt").encoding
'utf-8'
```

Learn More on realpython.com/search:

`utf-8` · `encoding` · `file handling`

Goodbyes

- Nothing headline-grabbing was removed
- The `sre_*` modules are gone, use `re`
- `profile` is deprecated, gone in 3.17
- `python -m http.server --cgi` is gone

Dates Now Need a Year

```
from datetime import datetime

datetime.strptime("03-14", "%m-%d")
# ValueError: Day of month directive
# '%d' may not be used without a
# year directive.
```

Learn More on realpython.com/search:

`deprecation` · `strptime` · `regex`

More 3.15 Highlights

What You Get	Where to Look
Integer-only math	`math.integer.isqrt(n)`
Clearer regex name	`re.prefixmatch(p, s)`
JSON arrays your way	`json.loads(s, array_hook=tuple)`
TOML 1.1 inline tables	`tomllib.loads(text)`
Consume a bytearray	`buffer.take_bytes()`
NaN-safe max and min	`math.fmax()`, `math.fmin()`
Cancel a task group	`asyncio.TaskGroup.cancel()`
More color in CLI tools	`python --help`, `sqlite3` shell
Upgraded experimental JIT	`PYTHON_JIT=1` on a JIT build